Daniel David

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Spoken Languages: English, Russian

Education

Innopolis University — Innopolis, Russia
MSc, Artificial Intelligence & Data Engineering — Aug. 2024 – Jul. 2026 | CGPA: 4.24

Tomsk Polytechnic University — Tomsk, Russia
BEng, Software Engineering — Sep. 2019 – Jul. 2024 | CGPA: 3.85

Experience

PRAAMS — Senior Software Developer — Sep. 2022 – Present

- Led architecture and development of PRAAMS, an AI-powered co-investor app combining conversational AI, stock analysis, and interactive financial visualizations.
- Conducted technical interviews and onboarded new team members.

Bellissimo Pizza — ML / Data Engineering Consultant — Jun. 2026 – Nov. 2026

- Built an end-to-end ML data pipeline for order-demand and price prediction using Apache Airflow, PostgreSQL, and MinIO; automated ingestion, transformation, and training-data preparation.
- Contributed to hourly demand forecasting for 50+ restaurants over 24/48/72-hour horizons, incorporating POS and weather features into model training and evaluation.
- Automated model retraining on new data and maintained data pipelines to support recurring forecasting workflows.

Innopolis University — Student Researcher — Aug. 2025 – Sep. 2026

- Implemented and trained ML/RL models in PyTorch; built web-scraping and big-data analysis pipelines using MongoDB, PySpark, and Apache Hadoop.

Selected Projects

nanoTitan — Open-Source Mixture-of-Experts Training Stack
Python, PyTorch, SDPA

- Added configurable SDPA / Flash Attention support to the existing multi-head attention module, retaining an eager attention path as a fallback.
- Improved the model architecture by integrating fused QKV projections and RMSNorm.

Multilingual Math Reasoning for Nigerian Languages
Docker, Postgres, Hugging Face, vLLM, MLX, GSM8K

- Built an ML data pipeline translating GSM8K questions and solutions into Igbo, Hausa, and Yoruba for Tiny Aya supervised fine-tuning on math solving and explanation; produced the gsm8k-lowres-math dataset.
- Benchmarked Qwen2.5-1.5B-Instruct and Tiny Aya families of models on math tasks across all three languages to guide model selection before fine-tuning.

ECG Benchmarking for Atrial Fibrillation Detection
PySpark, PostgreSQL, MinIO

- Contributed to a reproducible benchmarking framework spanning 42+ ECG databases, with deterministic data splits, cached feature generation, and traceable experiment artifacts.
- Helped benchmark 14+ detectors across datasets using window- and recording-level metrics, with quality and runtime comparisons through configurable experiments.

Selected Projects (continued)

Kinship Recognition from Photographs

ArcFace, SentenceTransformers, FAISS, CatBoost

- Contributed to dual-modal face/name retrieval on Families in the Wild, combining ArcFace and MiniLM embeddings through two FAISS indexes and reciprocal rank fusion.
- Worked on CatBoost classifiers for family membership and relationship type across 11 kinship categories, with SHAP-based explanations.

RAG for Document Collections

Python, SentenceTransformers, LLMs

- Contributed to building a Wikipedia-based RAG pipeline with article cleaning, chunking, 384-dimensional MiniLM embeddings, and incremental embedding updates.
- Combined 3-gram candidate retrieval with cosine-similarity reranking to supply relevant document snippets for LLM-generated answers.

Research

Meta-learners and LLM-Guided Adaptive Weighting for Physics-Informed Neural Networks

- Lead-authored a study introducing an LLM meta-controller that dynamically balances PDE-residual and boundary-condition losses using training history, validation trends, and stability diagnostics.
- Benchmarked LLM-guided weighting against fixed weights and LBPINN on Poiseuille flow across 100 pressure-gradient values and five random seeds, assessing convergence, solution accuracy, and adaptation trade-offs.

Recommendation Systems via Approximate Matrix Factorization

- Co-authored a project report comparing six matrix-factorization methods for collaborative filtering, evaluating RMSE, convergence, and sensitivity to rank and regularization.

Software Engineering Metrics Selection Using Equilibrium Optimizer and Differential Evolution

- Implemented the Equilibrium Optimizer for software-metric selection, using heuristic search to identify compact feature subsets while preserving geometric relationships between data points.
- Used Sammon error and Kruskal stress to evaluate structural preservation; the Equilibrium Optimizer reduced 30 metrics to five in a co-authored study also evaluating Differential Evolution.